Faculty of Computing and Technology
University of Kelaniya

GTEC 23032-Projects in Technology ||

<u>Project Report</u>

Project Title:- **EV Charging Point Locator**

Group Number:- 08

Supervisor:- Dr.Chanaka Udayanga

Group Leader:- CT/2021/087 – MARASINGHE M.M.C.U

Group members:- CT/2021/010 – CHAMARA B.P

CT/2021/033 - PERERA P.S.A

CT/2021/040 - LIYANAGE L.S.D.S.M

CT/2021/063 - DILSARA S.A.K.N

CT/2021/087 - MARASINGHE M.M.C.U

Date:-July 2025

Content

1. Introduction
2. Objectives
3. Literature Review
4. Methodology
5. System Overview
6. Results and Evaluation
7. SWOT Analysis
8. Conclusion
9. Future Work
10. References

Introduction

With the growing demand for electric vehicles (EVs) in Sri Lanka, the need for a reliable platform to locate charging stations has become increasingly important. As the country moves toward a greener and more sustainable future, electric mobility plays a crucial role in reducing carbon emissions and dependence on fossil fuels. However, one of the main barriers to EV adoption is the lack of accessible, real-time information on charging infrastructure.

International applications such as PlugShare and ChargePoint provide useful features, but they often fall short in addressing the specific needs of Sri Lankan users. These apps may not support local plug types, may lack station data in rural or emerging areas, and usually do not offer pricing in Sri Lankan Rupees. In addition, station owners have no control over adding or updating their station details in these platforms.

The E-Charging Point Locator app was developed to fill this gap. It provides EV users in Sri Lanka with real-time access to nearby charging stations using GPS, allowing them to filter by plug type and charging speed, view pricing, and make payments directly through the app. The platform also empowers station owners by allowing them to register and manage their station details. Overall, the app offers a localized, user-friendly, and scalable solution to support the country's EV infrastructure and promote the use of clean energy.

Objectives

- To develop a mobile application that displays nearby EV charging stations.
- To support various plug types and charging speeds.
- To allow charging station owners to register their stations.
- To include a simulated payment feature for demonstration purposes.
- To improve the EV user experience with accurate, real-time data.
- To encourage green transport by making charging infrastructure more accessible.
- To integrate GPS and mapping services for real-time location tracking.
- To support different user roles such as EV users and station owners.
- To enable filtering of stations based on plug type, speed, and price.
- To test usability and performance of the system in practical environments.

Literature Review

Existing apps like PlugShare, ChargePoint, Zap-Map, and EVgo provide solutions to locate EV charging stations worldwide. These apps typically offer maps, user ratings, and plug-type filters. However, they are often limited in their coverage of countries like Sri Lanka. For example, PlugShare and ChargePoint provide limited data in developing regions, and their interfaces and features are designed for Western markets.

Moreover, a study titled "Smart Management of EV Charging Stations" (2023) emphasizes the need for interactive and intelligent platforms that integrate location services and user-specific data. Another research work, "ChargeEV" (2024), introduces an app that includes real-time availability and reservation systems, but again lacks local customization. These findings suggest a gap in regionally focused, easy-to-use apps that empower local EV infrastructure.

Our project builds on these insights by offering a Sri Lanka-specific app with features like local plug compatibility, station owner input, and simulated payment integration. By focusing on user experience and local relevance, the E-Charging Point Locator aims to address shortcomings in global solutions.

Methodology

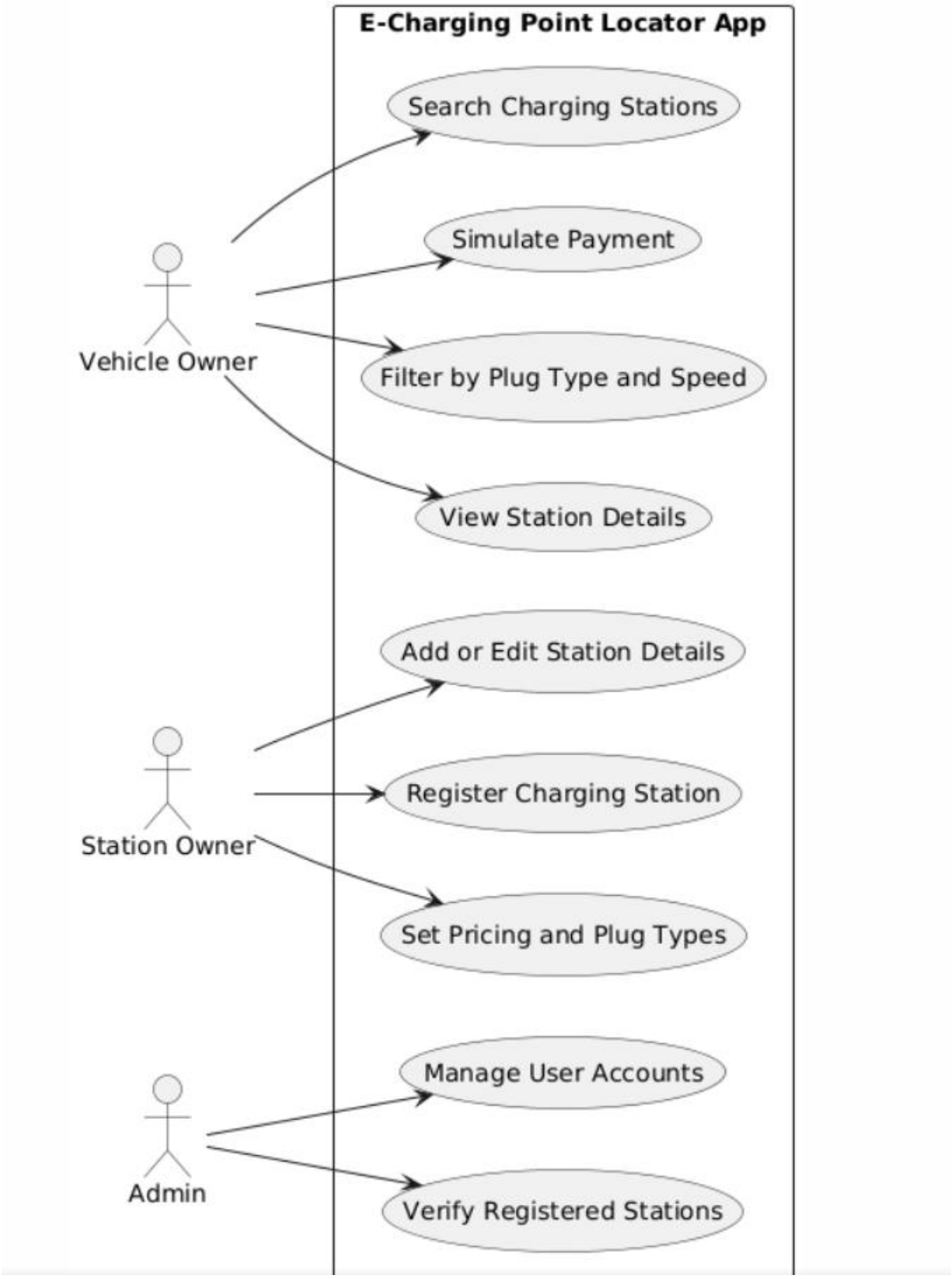
The development of the E-Charging Point Locator followed the Agile methodology, which allowed for iterative development and continuous feedback. The process was divided into the following stages:

- **Requirement Analysis:** Conducted surveys and informal interviews with EV users and station operators to understand pain points and desired features. Key requirements included real-time location, plug compatibility filters, and ease of use.
- **Design:** Created initial wireframes and prototypes using Figma. Designed intuitive user interfaces for both EV users and station owners. Focused on minimalistic and mobile-friendly layouts.
- **Development:**
    - **Frontend:** Developed using Flutter to support Android and cross-platform compatibility.
    - **Backend:** Utilized Firebase for real-time data handling, storage, and authentication.
    - **Mapping Services:** Integrated Google Maps API for accurate station visualization and user location tracking.
    - **Payment Simulation:** A dummy payment feature was implemented to simulate transaction flow for demonstration.
- **Testing:** Conducted unit testing on modules, user acceptance testing with sample users, and functional testing to ensure that each component works as expected.
- **Deployment:** Prepared for deployment within the Sri Lankan region. Though the current version uses static data input, future updates will automate data input and validation.

The methodology ensures the app is not only functional but also scalable and ready for future enhancements such as live booking and real-time charger availability.

System Overview

Use Case Diagram



Results and Evaluation

The E-Charging Point Locator app was tested internally by the development team using dummy station data to simulate real usage scenarios. Users were able to successfully locate nearby charging stations using GPS integration. The plug-type filtering system worked effectively, allowing filtering by plug types and speeds. The dummy payment simulation demonstrated how the actual payment process could work when integrated in the future.

Functionality tests focused on navigation, UI response time, and integration between user inputs and database retrievals. The app showed stable performance during test runs. Identified bugs were fixed, and final testing confirmed that the app meets its intended design goals.

SWOT Analysis

Strengths:

- Localized and customized for Sri Lanka's EV market.
- Real-time station locating using GPS.
- Flexible station registration by owners.
- Simple, clean UI designed for ease of use.
- Dummy payment simulation showcases system potential.

Weaknesses:

- Currently lacks live payment and booking system.
- Limited database of stations at launch.
- Dependent on manual data entry by station owners.

Opportunities:

- Can be expanded to support secure online payments.
- Potential to partner with CEB or private charging networks.
- Scope for multilingual support (Sinhala/Tamil).
- Could include station ratings, user reviews, and booking time slots.

Threats:

- Risk of global apps entering the local market with more features.
- Adoption may be slow if users are unaware of the app.
- Data integrity may suffer if station owners neglect updates.

Conclusion

The E-Charging Point Locator app offers a much-needed solution to Sri Lanka's growing EV ecosystem. By providing accurate and user-friendly access to charging station data, along with a simulated payment feature, the app supports sustainable transportation efforts. With future enhancements like real payment gateway integration, booking, and real-time station status, this project can continue to expand its impact and usability.

Future Work

- Replace dummy payment simulation with a secure live payment gateway.
- Add features such as booking integration and real-time station status.
- Expand station database through partnerships.
- Launch multi-language support for Sinhala and Tamil.
- Improve backend with smart data validation and analytics

References

- PlugShare App Overview
- ChargePoint Network Details
- Creswell, J. W. (2018). Research Design: Qualitative, Quantitative, and Mixed Methods Approaches. Sage Publications.
- Relevant academic and industry literature on EV infrastructure